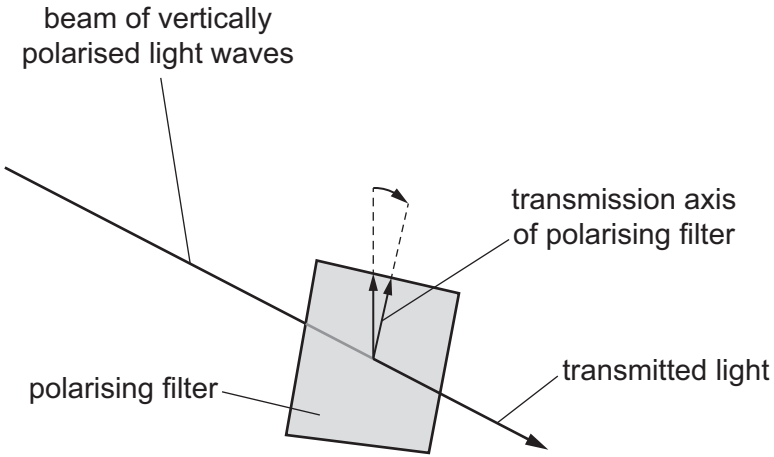
**24** A beam of vertically polarised light is incident normally on a polarising filter. The filter can be rotated so that it is always in a plane perpendicular to the beam. The transmission axis of the filter is initially vertical.




The filter is first rotated clockwise by an angle of 30° so that the transmitted light waves have intensity $I_{30}$.The filter is then rotated clockwise by a further angle of 30°.

What is the new intensity of the transmitted light waves?

**A** $0.25\,I_{30}$ **B** $0.33\,I_{30}$ **C** $0.75\,I_{30}$ **D** $0.87\,I_{30}$